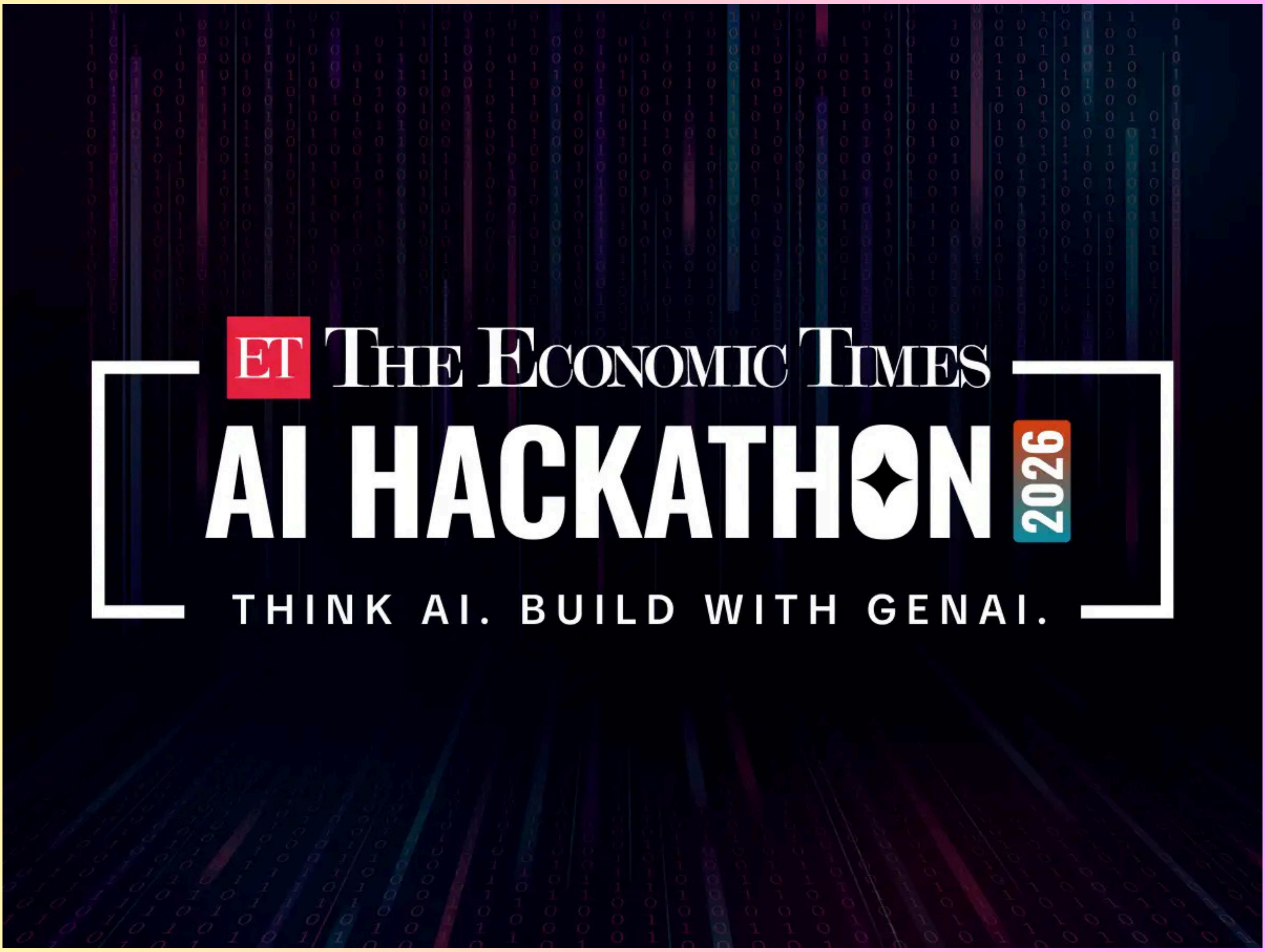
AGRIBOT - AI-POWERED FARMING ASSISTANT



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AgriBOT
AI-Powered Farming Assistant

ABSTRACT

Agriculture faces significant challenges due to unpredictable environmental conditions, lack of real-time advisory systems, and inefficient resource utilization. This paper presents AgriBOT, an AI-powered farming assistant that integrates machine learning and deep learning techniques to provide intelligent agricultural recommendations. The system performs crop recommendation, plant disease detection, and fertilizer suggestion using soil data and plant images. A convolutional neural network (CNN) based on MobileNetV2 is used for disease classification, while Random Forest algorithms are employed for crop and fertilizer prediction. The system is deployed through a user-friendly Streamlit interface, making it accessible and practical. Experimental results demonstrate that AgriBOT delivers accurate and actionable insights, contributing to improved agricultural productivity.

1. INTRODUCTION

Agriculture remains a critical sector worldwide, yet it is increasingly challenged by climate variability, soil degradation, and limited access to expert knowledge. Farmers often rely on traditional methods, which can lead to suboptimal crop selection, delayed disease identification, and improper fertilizer usage.

Recent advancements in Artificial Intelligence (AI) offer promising solutions to these challenges. By leveraging data-driven models, AI systems can assist farmers in making informed decisions. This paper introduces AgriBOT, a comprehensive AI-based system designed to provide real-time agricultural recommendations using multimodal inputs such as soil parameters and plant images.

2. LITERATURE REVIEW

Several studies have explored the application of AI in agriculture. Machine learning models such as Decision Trees, Support Vector Machines, and Random Forests have been widely used for crop recommendation based on soil properties. Deep learning models, particularly Convolutional Neural Networks (CNNs), have shown high accuracy in plant disease detection tasks.

The PlantVillage dataset has been extensively used for disease classification, achieving significant success with transfer learning approaches such as MobileNet and ResNet. However, most existing systems focus on a single task, such as disease detection or crop recommendation, rather than providing a unified solution.

AgriBOT addresses this limitation by integrating multiple AI models into a single pipeline.

3. SYSTEM ARCHITECTURE

The architecture of AgriBOT consists of three main modules -

1. Disease Detection Module (Deep Learning).
2. Crop Recommendation Module (Machine Learning).
3. Fertilizer Recommendation Module (Machine Learning).

Workflow -

1. Input - Leaf image and soil parameters.
2. Disease Detection using CNN.
3. Crop Prediction using soil features.
4. Fertilizer Recommendation using crop and soil data.
5. Output - Combined advisory.

4. METHODOLOGY

4.1 Dataset Preparation

Disease Dataset

The PlantVillage dataset was used for disease detection. The original dataset contained nested directories (Train, Validation, Test), which were flattened into a single-level structure - "Crop___Disease".

This transformation was necessary for compatibility with deep learning data loaders.

4.2 Data Preprocessing

- Image resizing to 128×128 pixels.
- Normalization of pixel values (0–1 range).
- Label encoding for categorical variables in fertilizer dataset.
- Feature alignment for consistent model input.

4.3 Model Development

4.3.1 Disease Detection Model

- Architecture - MobileNetV2 (Transfer Learning).
- Loss Function - Categorical Crossentropy.
- Optimizer - Adam.

MobileNetV2 was selected due to its lightweight nature and high performance in image classification tasks.

4.3.2 Crop Recommendation Model

- Algorithm - Random Forest Classifier.
- Input - Soil parameters (N, P, K, temperature, humidity, pH, rainfall).

Random Forest was chosen for its robustness and ability to handle non-linear relationships.

4.3.3 Fertilizer Recommendation Model

- Algorithm - Random Forest Classifier.
- Input - Soil data + Crop type.

Categorical variables were encoded using LabelEncoder, and feature consistency was maintained using stored column mappings.

4.4 Prediction Pipeline

The system follows a multi-stage prediction pipeline -

1. Image → Disease Detection.
2. Soil Data → Crop Recommendation.
3. Crop + Soil → Fertilizer Recommendation.

5. IMPLEMENTATION

The system was implemented using -

- Python.
- TensorFlow / Keras.
- Scikit-learn.
- Streamlit.

A web-based interface was developed using Streamlit, enabling users to interact with the system through an intuitive UI.

6. AGENT ROLES AND RESPONSIBILITIES

6.1 Disease Detection Agent

- Type - Deep Learning (CNN – MobileNetV2).
- Input - Leaf image.
- Output - Crop + Disease.

Identifies plant disease from visual symptoms.
Splits label into “Crop___Disease”.

6.2 Crop Recommendation Agent

- Type - Machine Learning (Random Forest).
- Input - Soil parameters (N, P, K, temperature, humidity, pH, rainfall).
- Output - Recommended crop.

Provides optimal crop selection.
Works independently of image input.

6.3 Fertilizer Recommendation Agent

- Type - Machine Learning (Random Forest).
- Input - Crop + Soil data.
- Output - Fertilizer suggestion

Uses encoded categorical features.
Ensures correct feature alignment.

7. COMMUNICATION FLOW

The agents communicate through a sequential pipeline -

1. User uploads image + inputs soil data.
2. Disease Agent processes image.
3. Crop Agent processes soil data.
4. Outputs are passed to Decision Agent.
5. Fertilizer Agent uses combined data.
6. Final output is generated and displayed.

8. RESULTS AND DISCUSSION

The system successfully performs -

- Accurate disease detection from leaf images.
- Reliable crop recommendation based on soil conditions.
- Effective fertilizer suggestion.

Sample Output -
Detected Crop - Tomato
Disease - Late Blight
Recommended Crop - Rice
Fertilizer - Urea
The integration of multiple models improves the overall decision-making process, making the system more practical for real-world use.

9. CHALLENGES AND SOLUTIONS

Dataset Structure Issue

- Problem - Nested dataset prevented proper labeling.
- Solution - Flattened dataset into “Crop___Disease” format.

Feature Mismatch in Fertilizer Model

- Problem - Inconsistent input features.
- Solution - Maintained feature order using saved column metadata.

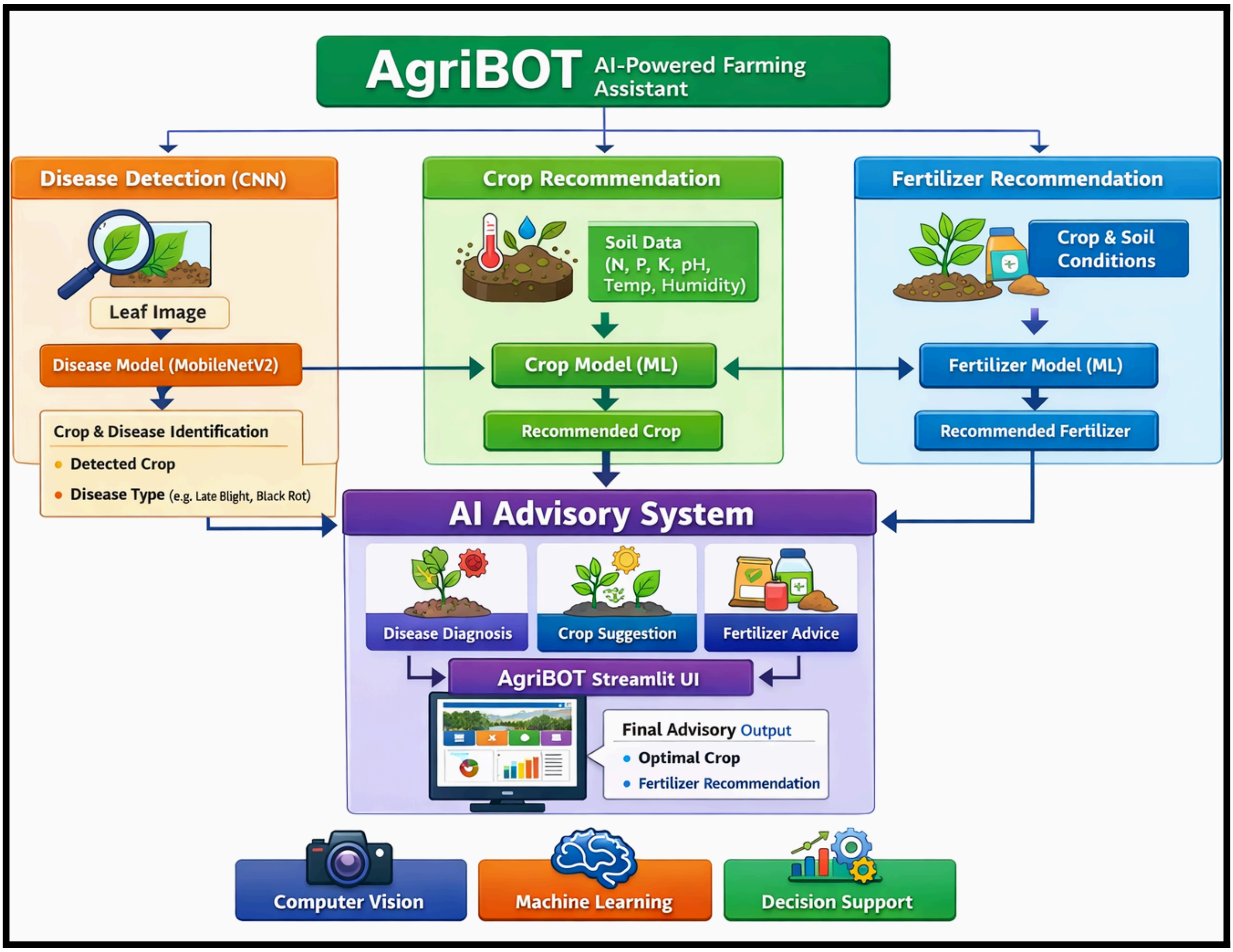
Deployment Issues

- Problem - Streamlit and ngrok connectivity errors.
- Solution - Correct execution sequence and debugging.

10. FUTURE WORK

- Integration of real-time weather APIs.
- Voice-based interaction in local languages.
- SMS-based advisory system.
- Market price prediction module.
- Integration of Large Language Models (LLMs) for reasoning.

11. SYSTEM ARCHITECTURE



12. CONCLUSION

AgriBOT demonstrates the potential of AI in transforming agriculture by providing intelligent, data-driven recommendations. By integrating machine learning and deep learning models, the system offers a comprehensive solution for crop selection, disease detection, and fertilizer management.

The system is scalable, user-friendly, and adaptable for real-world deployment, making it a valuable tool for modern agriculture.

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